Feature Importance for Applied Research

From model inspection to statistical inference

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- *Sanity* check: does a model rely on plausible signal?

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> **!** “Importance” is not a single quantity → *method choice* defines the answer

Which features matter?

Term	Estimate	SE	p
`hour`	7.8	0.4	< 0.001
`humidity`	−197.6	16.4	< 0.001
`temperature`	350.8	22.5	< 0.001
`windspeed`	18.9	22.6	0.403

`lm(count ~ .)` on bike sharing data (more later)

- Linear model: compare $|\hat{\beta}_j|$, t-test p-values

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But: Important *for what*, and *how* is it measured?

Importance ≠ effect

Effect

- "What happens to $\hat{y}$ *when* X_j *changes*?"
- Linear model: β_j
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  - Accumulated local effects (*ALE*)

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- Scale depends on the loss → read as a *ranking*

Taxonomy & Estimands

A rough taxonomy of interpretibility methods

Today: model-agnostic, loss-based, global

Model-specific

(G)LM coefficients, Gini impurity, gradient-based

|

vs.

|

Model-agnostic

supports any learner

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Local
One observation’s prediction (e.g., SHAP)

vs.

Global
the model as a whole, feature-specific

What are we estimating?

Which question?

- **Marginal** ("crude"): $X_j \not\perp\!\!\!\perp Y$

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- **Learner**: the fitting procedure, across datasets
- **Data-generating process** (Williamson et al. 2023)
  - Learner importance + Bayes-optimal predictor

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⚠ Most methods estimate *model* importance, research questions usually target the *data*

Feature Importance Methods

Why not just refit?

Leave-one-covariate-out (LOCO) (Lei et al. 2018)

Calculation

1. Fit model $\hat{f}$ with all features
2. Refit $\hat{f}_{-j}$ without X_j
3. Compare test loss

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- Groups: leave-one-*group*-out (LOGO)

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- Effectively: one step of backward selection
- Groups: leave-one-*group*-out (LOGO)
- Leave-one-covariate-*in* (LOCI) = univariate association
- *Cost*: one refit per feature → expensive for
  - Complex algorithms / tuning
  - Huge datasets

📝 **Test loss, e.g. mean squared error**

$L\left(\hat{f}\right) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}(x_i)\right)^2$ → importance is on the scale of the *loss*

Don’t refit, perturb

Permutation feature importance (PFI) (Breiman 2001)

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1. Fit model $\hat{f}$ once
2. Shuffle X_j in the test data: $\tilde{x}_j$
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4. Repeat k times for stability

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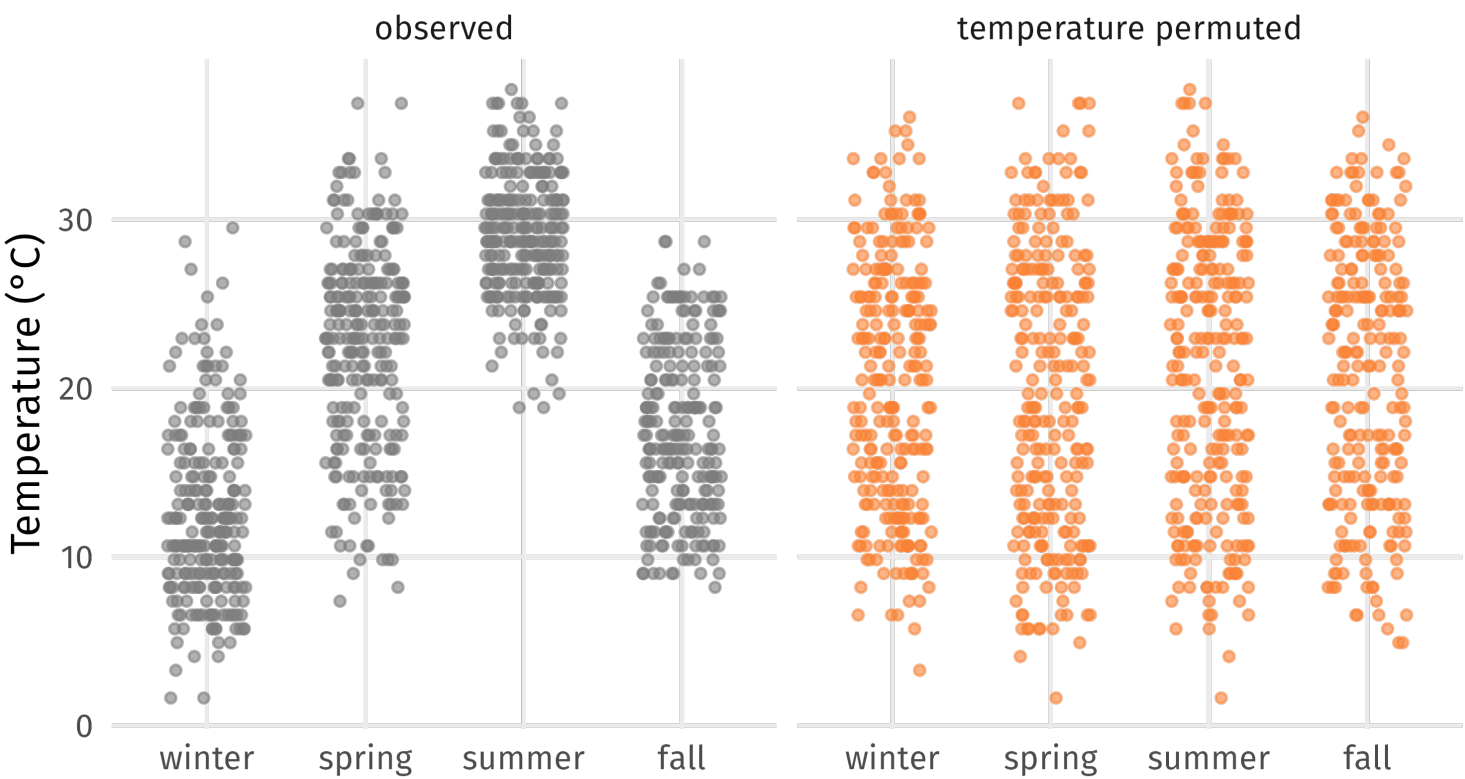
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- Breaks the link to Y *and* to the other features
- Correlated features → implausible combinations
  - “Off-manifold” problem / forced extrapolation
  - → A 20-year-old patient with 30 years of smoking history

Shuffling leaves the data

Model is evaluated where it has never seen data (Hooker et al. 2021)



Perturb, but keep the dependence

Conditional feature importance (CFI)

- Sample replacement values $\tilde{x}_j$ from $X_j \mid X_{-j}$ instead of marginal shuffling
- Removes only the part of X_j *not* explained by the others
- Stays on the data manifold

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- Otherwise: Same procedure as PFI:

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! Needs samples from $X_j \mid X_{-j}$, which are not easy to get

Between PFI and CFI

Relative feature importance (RFI) (König et al. 2021)

Sample X_j from $X_j \mid X_G$ for a chosen *conditioning set* G

$G = \varnothing$

- PFI
- "Does the model use X_j?"

$G = \{\text{age}, \text{sex}\}$

- RFI
- "Beyond age and sex?"

$G = -j$

- CFI
- "Beyond all other features?"

Where do conditional samples come from?

- *Gaussian*: fast, continuous features only
- *Knockoff* framework (Candès et al. 2018)
- Tabular data is usually *mixed*: continuous, categorical, missing values

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📝 **Adversarial random forests (ARF)**

- Generative model for tabular data, estimates conditional density (Watson et al. 2023)
- Mixed types, missing values
- Default sampler for CFI and RFI in `{xplainfi}` (Blesch et al. 2025)

Is it more than noise?

Inference for CFI

- CFI is a mean of per-observation loss differences, $\tilde{x}_i$: x_i with x_{ij} perturbed

$$\Delta_i = \left(y_i - \hat{f}(\tilde{x}_i)\right)^2 - \left(y_i - \hat{f}(x_i)\right)^2, \quad \text{CFI}_j = \overline{\Delta} = \frac{1}{n}\sum_{i=1}^{n} \Delta_i$$

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- One-sided paired test on $\overline{\Delta}$ (e.g., t-test, Wilcoxon)
  - *CPI*: "conditional predictive impact" test using knockoffs (Watson and Wright 2021)
  - *cARFi*: same test with ARF sampling (Blesch et al. 2025)

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💡 Holdout set, model fixed
→ *model* importance

📝 Many features:
Control FWER or FDR
(Holm, Benjamini–Hochberg, ...)

And now for some game theory

Shapley Additive Global importancE (SAGE) (Covert et al. 2020)

- Value of a feature set S (“coalition”):

  Expected loss reduction when only features in S are known

$$v(S) = \mathbb{E}[\ell(Y, f_\varnothing)] - \mathbb{E}[\ell(Y, f_S(X_S))]$$

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$$\varphi_j = \sum_{S \subseteq D \setminus \{j\}} \frac{|S|!(p - |S| - 1)!}{p!} [v(S \cup \{j\}) - v(S)]$$

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📝 **SAGE vs. SHAP**

SAGE: *global*, explains the loss
(Covert et al. 2020)

SHAP: *local*, explains one prediction
(Lundberg and Lee 2017)

Application Example

The bike sharing dataset

Capital Bikeshare, Washington, D.C., 2011–2012

- *Outcome*: bike rentals per hour
- 3000 hours sampled from ~17k
- Gradient boosting (`xgboost`, pre-tuned) and `lm`
- Holdout: 80% train, 20% test

9 features

- **Calendar**: hour, weekday, working day, season, year
- **Weather**: temperature, humidity, wind speed, weather type

Setup for code example

{mlr3} objects, defined once


```
library(xplainfi)
library(mlr3learners)
library(mlr3pipelines)

task <- tsk("bike_sharing")
learner <- as_learner(
  po("encode") %>>%
    lrn("regr.xgboost", nrounds = 4932, eta = 0.0023, max_depth = 8)
)
measure <- msr("regr.mse")
resampling <- rsmp("holdout", ratio = 0.8)
```

PFI


```
pfi <- PFI$new(
  task, learner, measure,
  resampling = resampling,
  n_repeats = 10
)
pfi$compute()
pfi$importance()
```

Feature	PFI
hour	43478
temperature	8526
working_day	6532
year	6120
humidity	1983
season	1365
weekday	912
weather	673
windspeed	98

CFI


```
cfi <- CFI$new(
  task, learner, measure,
  resampling = resampling,
  n_repeats = 10,
  sampler = ConditionalARFSampler$new(task)
)
cfi$compute()
cfi$importance()
```

Feature	CFI
hour	36061
year	5537
temperature	4817
working_day	3268
humidity	1180
season	1020
weekday	735
weather	567
windspeed	89

LOCO


```
loco <- LOCO$new(
  task, learner, measure,
  resampling = resampling
)
loco$compute()
loco$importance()
```

Feature	LOCO
hour	20460
year	3285
temperature	1253
working_day	646
season	542
weather	331
weekday	239
humidity	150
windspeed	-32

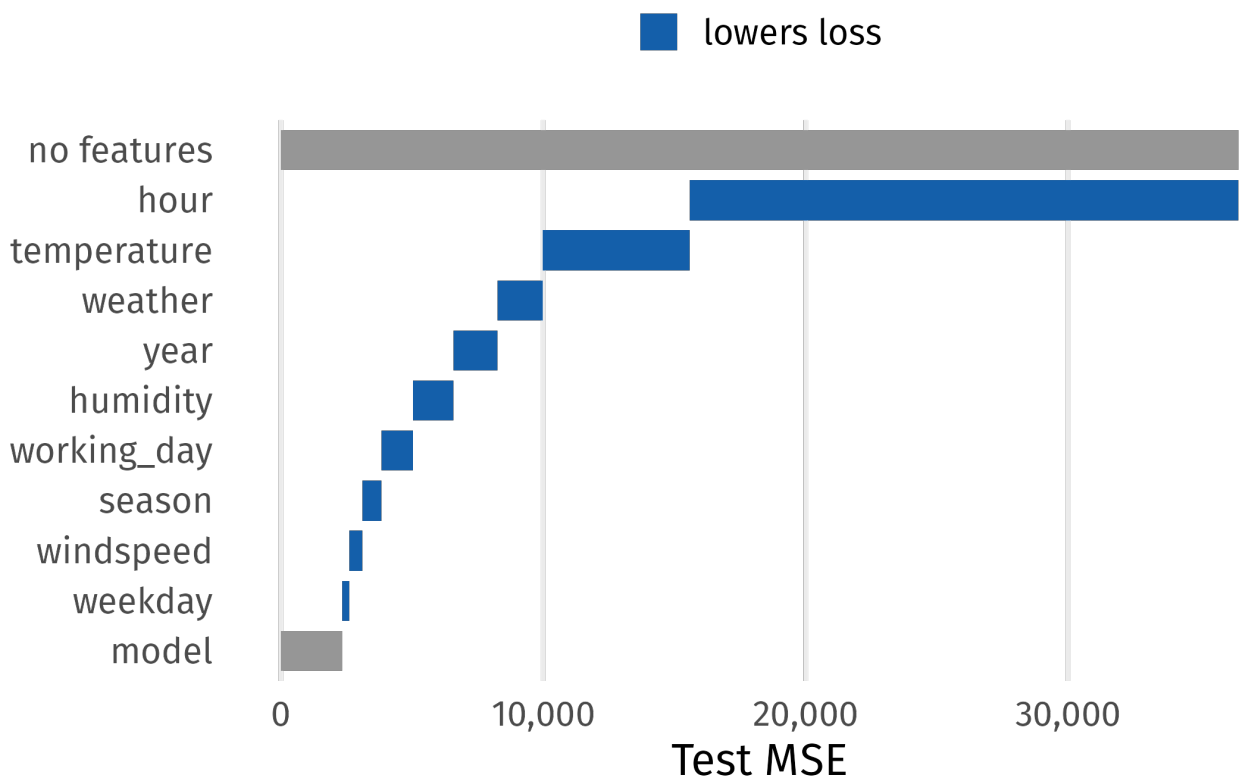
SAGE


```
sage <- MarginalSAGE$new(
  task, learner, measure,
  resampling = resampling,
  estimator = "kernel",
  n_coalitions = 16384
)
sage$compute()
sage$importance()
```

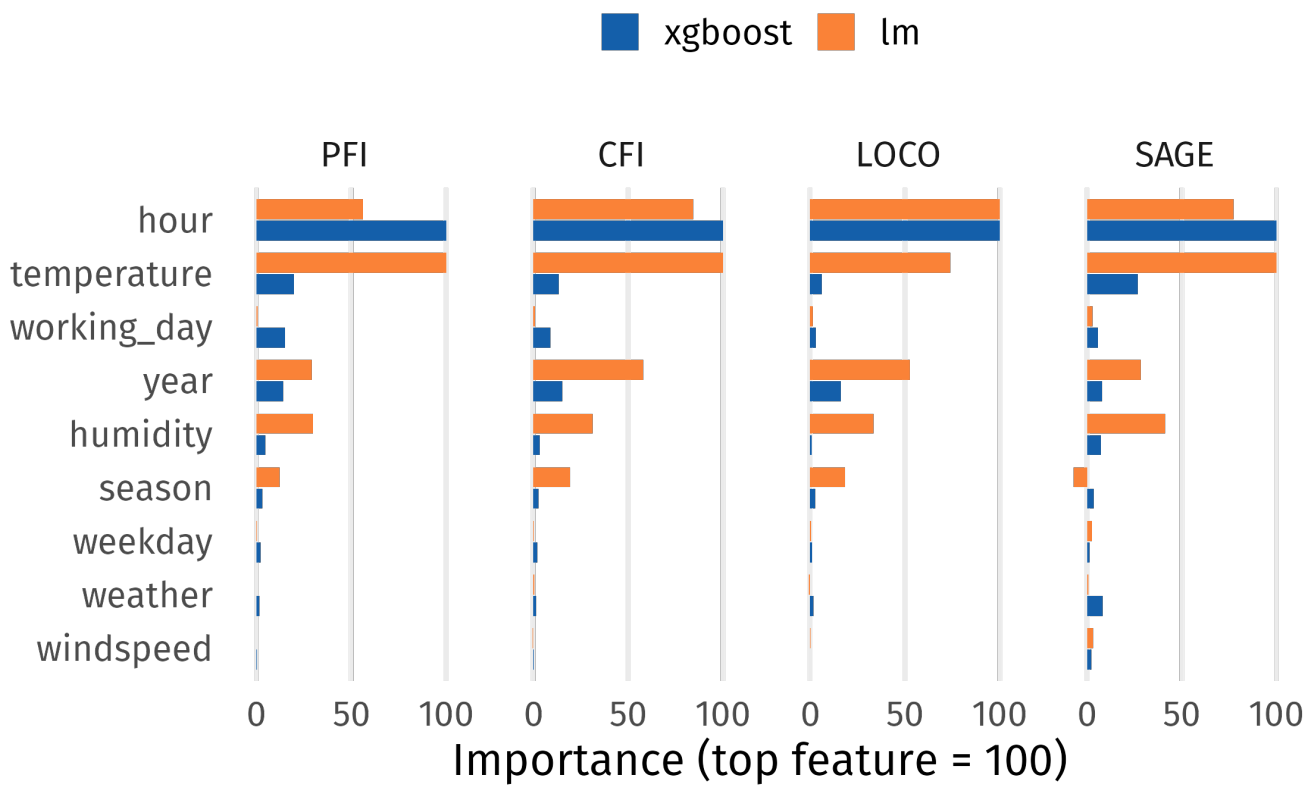
Feature	SAGE
hour	20923
temperature	5593
weather	1731
year	1687
humidity	1533
working_day	1197
season	725
windspeed	499
weekday	274

SAGE adds up

Loss without features → loss of the model, one step per feature



Comparing methods and learners



Inference

CFI with CPI, 600 held-out hours


```
cfi$importance(ci_method = "cpi", p_adjust = "BH")
```

Feature	CFI	SE	Lower 95%	p (BH)
hour	36061	2035	32064	< 0.001
year	5537	491	4572	< 0.001
temperature	4817	541	3754	< 0.001
working_day	3268	349	2582	< 0.001
humidity	1180	239	710	< 0.001
season	1020	146	733	< 0.001
weekday	735	99	540	< 0.001
weather	567	122	328	< 0.001
windspeed	89	66	-40	0.175

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- `hour`, `year`, `temperature` clearly important
- `windspeed`: no evidence after BH adjustment

Summary

Question	Method	Watch out for
Does the model use X_j?	PFI	extrapolation
Information beyond the others?	CFI: perturb	conditional sampler
	LOCO: refit	refits, learner tuning
Share of the explained loss?	SAGE	cost, estimator noise

- CFI and LOCO: near-identical estimands, different estimation (Ewald et al. 2024)

! Pick the method by the question, then check the estimand

Where to go next

1. `install.packages("xplainfi")`
2. Documentation: https://mlr-org.github.io/xplainfi
3. `xplainfi` and practical guidance:

   xplainfi: Feature Importance and Statistical Inference for Machine Learning in R

   Burk et al. (2026), arXiv:2603.15306
4. Methodological background:

   A Guide to Feature Importance Methods for Scientific Inference

   Ewald et al. (2024), Explainable Artificial Intelligence (xAI 2024)

Thank you for your attention!

Contact

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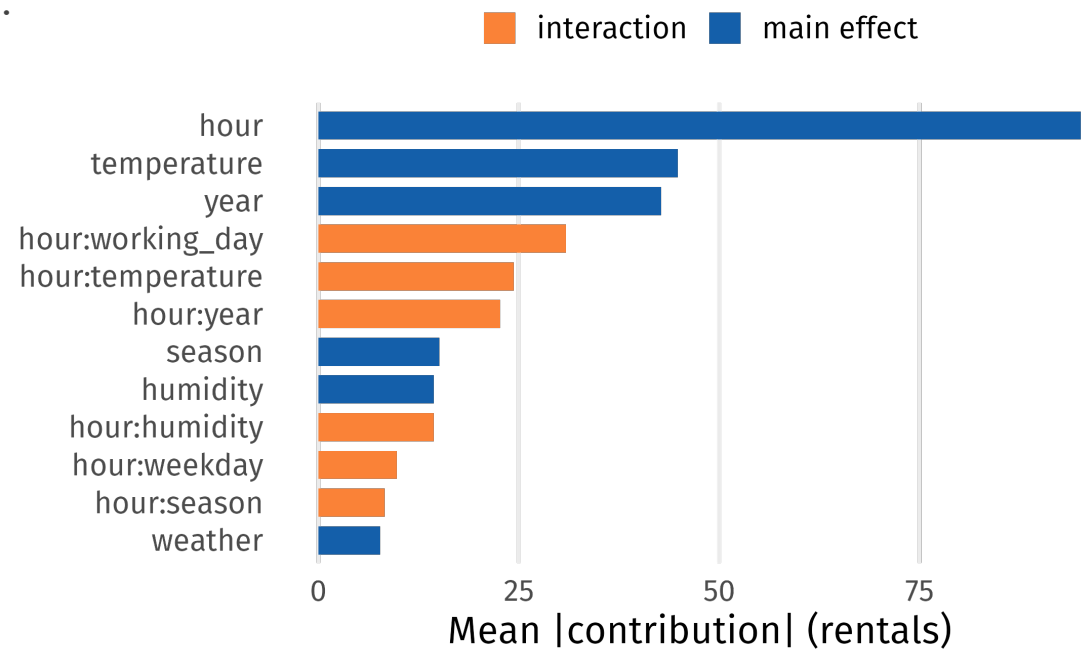
Backup

Prediction-based: decomposing $\hat{f}$

Functional decomposition of the `xgboost` model with `glex` (Hiabu et al. 2023)

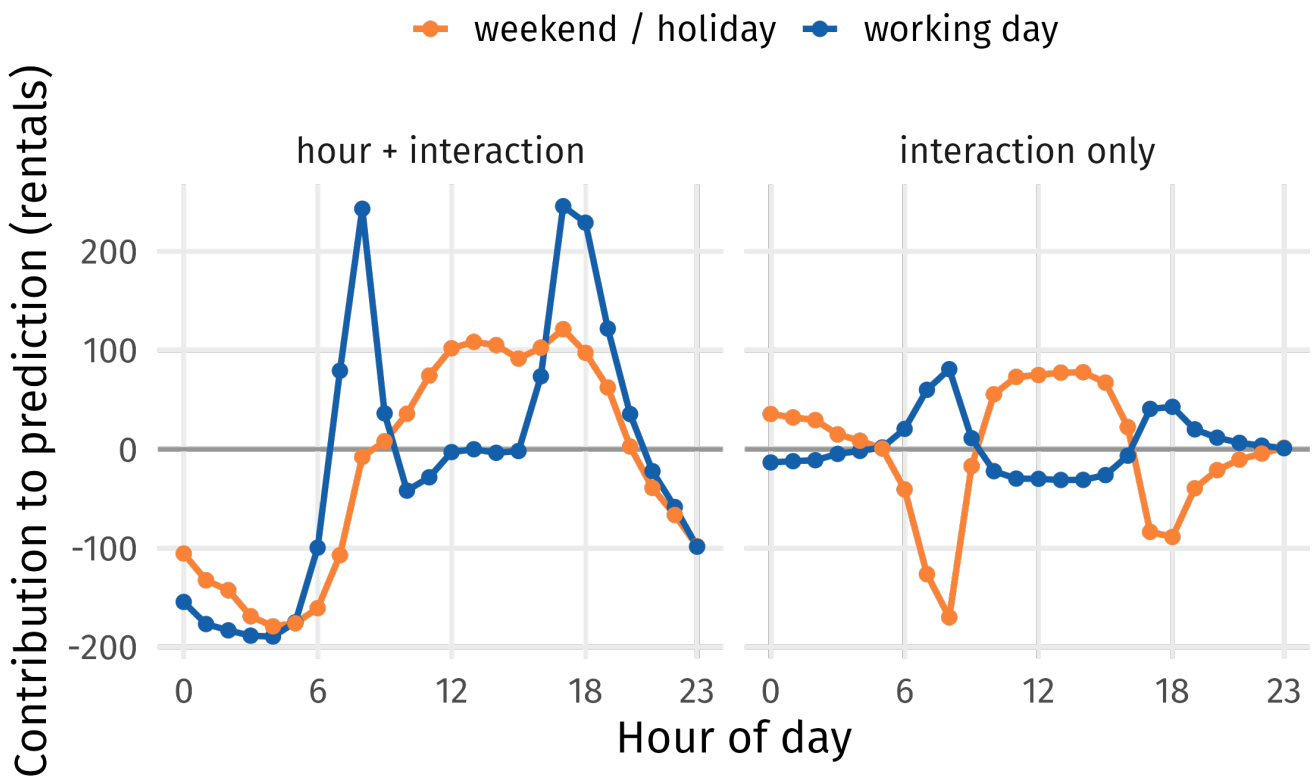
$$\hat{f}(x) = m_0 + \sum_j m_j(x_j) + \sum_{j<k} m_{jk}(x_j, x_k) + \dots$$

- Components on the scale of the *prediction* (rentals), not the loss
- Importance of a component: mean $|m_S|$
- No refits, no sampling: read off the trees



Low importance ≠ no effect

`working_day` matters through its interaction with `hour`



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